PANINI LINGUISTICS OLYMPIAD

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MEDIUM, HARD & EXPERT PROBLEMS

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A Polished Weather Forecast

You are starting a job as an IT specialist in a weather station. To handle the large number of forecasts every day, the station uses the Polish logical notation[1] to store forecasts on their computer systems in a concise and unambiguous manner. To work with these systems, it is essential that you understand how the notation works.

In this notation, simple logical statements such as 'it will rain today' are assigned variables and are connected by logical operators, just like how variables and mathematical operators are used in maths to construct algebraic expressions and equations like *2p/q* or *4x + 5y = 7*.

Here, you are given a list of weather forecasts in English, followed by their 'translations' in the logical notation in **random order**. Variables are arbitrarily assigned to logical statements. Keep in mind that *or* is inclusive throughout this exercise (i.e. x *or* y means x *or* y *or both*).

1. It will rain today but not tomorrow.
2. If it rains today or there is both cold weather today and no snow tomorrow, it will be warm tomorrow.
3. If it rains today and tomorrow, it will be cold tomorrow but it will not snow.
4. It will not snow both today and tomorrow (but it might snow one of the days).
5. If it rains tomorrow, either it will not snow tomorrow or it will rain today.
6. It will be warm today or it will be cold and either snow or rain tomorrow.
7. There will be no snow today or tomorrow.
8. It is either warm today or it will neither rain today nor tomorrow.
9. If it rains or snows today, there will be cold weather tomorrow.
10. It will not be cold today.
11. It will be cold today and if it doesn't rain today, it will snow tomorrow.

(a) `AsNApw` (b) `AsKqAtw` (c) `CApuq` (d) `CApKvNtr` (e) `Nv` (f) `CKpwKqNt`
(g) `NKut` (h) `KvCNpt` (i) `KpNw` (j) `CwANtp` (k) `NAut`

Assignment 1:

Identify the correspondences between the forecasts (1-11) and their representations in logic notation (a-k).

[1] Polish notation (PN), also known as normal Polish notation (NPN), Łukasiewicz notation, Warsaw notation, Polish prefix notation or simply prefix notation, is a mathematical notation in which operators precede their operands. When Polish notation is used as a syntax for mathematical expressions by programming language interpreters, it is readily parsed into abstract syntax trees. Because of this, Lisp and related programming languages define their entire syntax in prefix notation.

Assignment 2:

Convert these statements in logic notation to weather forecasts in English:

a. `KNpq`

b. `CAuNrw`

c. `KNsCNpw`

Assignment 3:

Convert these weather forecasts into logic notation.

(a) Either it will be neither cold nor warm tomorrow, or it will rain tomorrow.

(b) It will snow today and if it is cold tomorrow, it will snow tomorrow.

(c) If it rains today but not tomorrow, it will not be cold either today or tomorrow (but it could be cold one of the days).

Assignment 4:

Explain briefly, in your own words, what the following symbols are used for in the logical notation:

(1) `A` (2) `C` (3) `K` (4) `N`

– Ishan Ganguly

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

Problem no 3. Tangkhul Verbs

The following data is based on the verb forms in Tangkhul.

Tangkhul is a language belonging to the Sino-Tibetan language family and is spoken by Tangkhul-Naga community in Manipur, India.

Little Akham, a Tangkhul-speaking girl, 5 years of age, is chatting with her grandfather who has come to stay with her after many days. Read their dialogue carefully, and fill in the blanks with appropriate forms of the Tangkhul verbs given in brackets. If the verbs are not given in the brackets, they must have occurred in the dialogue before. Find the verbs and fill in their appropriate forms.

Akham: Aavaa bazaarli vaa hayra (Father has gone to the market). Aavu aina i ngatsinshon vaara (Mother and I will go in the evening.) Ithum haptakachida bazaarli vaaya (We go to the market every week). Achi i ice-cream-shopli vaara (Today I will go to the icecreamshop). I thangthangpingwi shopli vaangaaya (I want to go to the shop of toys).

I ice-cream_____(eat = sha) (I will eat ice-cream). I hainathaiwi ice-cream_____(eat =sha) (I have eaten mango ice-cream). I strawberrywi ice-cream_____(eat = sha) (I want to eat strawberry ice-cream). I thangthangping lora (I will buy toys).

Grandfather: Hi kaphaa na! (This is good!) I nali raaka (Let me come with you). I nali bazaarli raara (I will come with you to the market). I aathaiyur _____ (I want to buy fruits).

_____(Let me eat ice-cream).

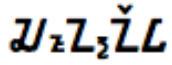
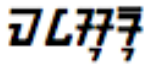
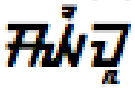
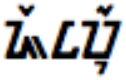
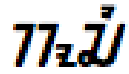
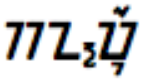
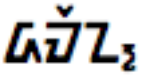
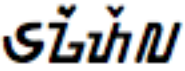
Akham: Kha kassariaping ice-cream mashashap mara (But, old people cannot eat ice- cream). Athum rria! (They are old!).

Grandfather: Chi-aina, i thangthangping_____(Then I will buy toys).

Akham: Kha, kassariaping mangaryao mana (But old people do not play). Na ithumli bazaarli_____(You cannot come with us to the market). Naaoshinaaoping aina athumwi aavu-aavaa bazaarli vaashapra (The children and their parents can go to the market.

Problem 2: Aksara Sunda Kuna (15 marks)

Here is a list of names of Indian cities transcribed in the *Sundanese* script, along with their *English* transliterations given in random order.

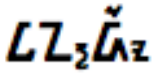
Sundanese	**English**
A. 	
B. 	*1. Mysuru*
C. 	*2. Jaipur*
D. 	*3. Agartala*
E. 	*4. Chennai*
F. 	*5. Kochi*
G. 	*6. Daman*
H. 	*7. Delhi*
I. 	*8. Shimla*
J. 	*9. Kanpur*
	10. Agra

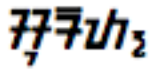
Assignment 1:

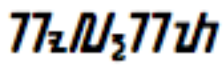
Find the correspondences between the names of cities in the *Sundanese* script and *English* script. Explain your answers. Write only the number corresponding to the Roman transliteration next to the letter corresponding to the *Sundanese* transliteration in your Answer Booklet. (For example: K - 11)

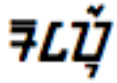
Assignment 2:

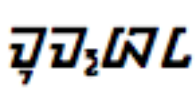
Here are the names of some more cities in the *Sundanese* script. Which cities are they?
If you cannot figure out the name of a city from the given data, write "*NP*" in your Answer Booklet, and then explain your reasoning.

a) 

b) 

c) 

d) 

e) 

Aksara Sunda Kuna is the written script of *Sundanese*, a Malayo-Polynesian language spoken by about 39 million people mainly in Western Java in Indonesia. It is the third most-spoken language in Indonesia. *Sundanese* is closely related to *Madurese* and *Malay*, and more distantly related to *Javanese*.

- Saujas Vaduguru

Problem 1: Brill's Tagger (15 marks)

A part-of-speech (POS) tagger is a piece of software that takes a sentence as input and outputs a tag for each word indicating the part of speech: broadly speaking, noun, verb, preposition, etc. One of the first statistical taggers ('statistical' meaning it was trained on sample data) was developed by Eric Brill in 1995 and is popularly known as Brill's Tagger. This worked in two steps. In the first step, each word was assigned the most likely tag, irrespective of context. For example, since 'book' as a noun is more common than 'book' as a verb, the first step would always mark the word 'book' in any sentence as a noun (symbol `NN`). Then, in the second step, some rules are applied to change the tags of some words based on context.

In the tables below, the first row has the original sentences, the second row has the result of the first step (assigning tags), and the third row has the result of the second step (applying rules). Note that some of the tags have been omitted. You may be able to figure out the meanings of some of the tags, but that is not required to solve this problem.

(a)

Saujas	books	one	ticket	for	the	show	.
NNP	NNS	CD	NN	PP	DT	VF	PU
NNP	**(1)**	CD	NN	PP	DT	NN	PU

(b)

Since	the	show	is	on	for	two	days	,
PP	DT	VF	VF	PP	**(2)**	CD	NNS	PU
CNJ	DT	NN	VF	PT	PP	CD	NNS	PU

he	could	go	on	both	days	.
PRP	VM	VF	PP	DT	NNS	PU
PRP	VM	VF	PP	DT	**(3)**	PU

(c)

Saujas	watches	it	on	the	first	day	.
NNP	NNS	PT	PP	DT	OD	NN	PU
NNP	VF	**(4)**	PP	DT	OD	NN	PU

(d)

He	suggests	Shardul	to	watch	it	as	well	.
PRP	**(5)**	NNP	PT	VF	PT	RB	**(6)**	PU
PRP	VF	NNP	**(7)**	VF	**(8)**	RB	RB	PU

(e)

Shardul	books	another	ticket	and	watches	it	the	next	day	.
NNP	NNS	DT	NN	CNJ	NNS	**(9)**	DT	JJ	NN	PU
NNP	VF	DT	NN	CNJ	RB	**(10)**	DT	JJ	NN	PU

The following rules are applied one by one in the given order. Note that some parts of the rules have been omitted.

(A) PT PRP WDPREVTAG VF it

(B) PP **(11)** **(12)** DT since

(C) PP PT WDPREVTAG VF WDNEXTTAG PP

(D) NNS VF **(13)** NNP

(E) VF NN WDPREVTAG **(14)**

(F) NNS **(15)** WDNEXTTAG PT

Assignment 1: (12 marks)

Fill in the omitted entries numbered (1)–(15). It may be helpful to know that rule (C) affects only one word and rule (E) does not affect sentence (c).

Assignment 2: (3 marks)

What will Brill's Tagger output for this sentence in both steps?

"He books it for another show."

– Monojit Choudhury

Problem No. 6 (25 Points):
Malagasy

Consider the following Malagasy words and their **unordered** English translations:

mahandohalika	grandson
lohalika	ankle
zafin-kitrokely	shoot of rice (departing from the stem)
hafaladia	up to the sole
zafim-bary	rice field
kitrokely	grand grand grand grandson
zafim-paladia	one who can get on his knees
zafy	grand grand grand grand grandson
tanim-bary	knee
mahambozona	one who can carry something on his neck

Assignment 1. Pair up the Malagasy words with their correct English translations. Explain your solution.

Assignment 2. Translate into English:

tany, vozona, halohalika

Assignment 3. Translate into Malagasy:

grand grand grandson, sole, up to the ankle

Note. *y* is pronounced as *i*.

[1] The Malagasy language belongs to the Austronesian language family and is the national language of Madagascar. It is spoken by more than 10 mln people.

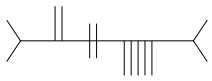
- A. Kretov

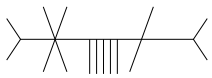
Problem 3: Ire-lined (11 marks)

Ogham is an Early Medieval writing system used for the Ancient (Early) Irish language and later the Old Irish language. There are roughly four hundred surviving inscriptions using Ogham (the so-called "orthodox inscriptions") on stone monuments throughout Ireland and western Britain.

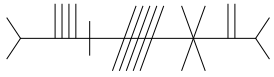
Assignment 1: (4 marks)

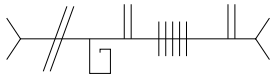
Match these Ogham (O) transcriptions to their Ancient Irish (AI) counterparts and their English (E) meanings. (Note: There is one fewer O option than there are AI and E options. The extra AI and E match up to each other, and to one O option from Assignment 2.)

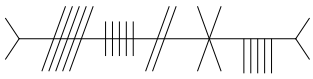
A. 

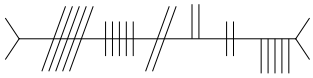
B. 

C. 

D. 

E. 

F. 

G. 

H. 

I. 

J. 

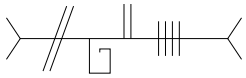
K. 

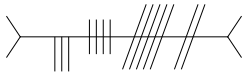
rigdon, lobraigidir, guidid, caratrad, erassaigidir, ainm, rigain, caraid,
ferg, lobur, don, aimnaigidir

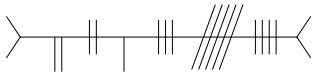
(somebody) names, queen, angry, (somebody) weakens, friendship,
royal palace, loves continuously, name, fort, (somebody) rejects, weak,
prays continuously

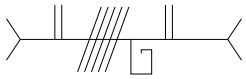
Assignment 2: (5 marks)

Of the O forms below, which one means 'priest'? What are the corresponding AI and E forms for all of them?

(a) 

(b) 

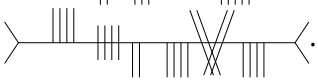
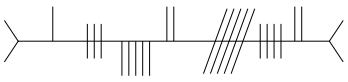
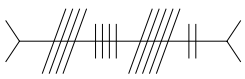
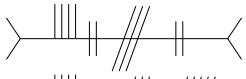
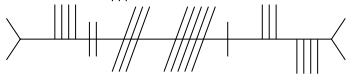
(c) 

(d) 

(e) 

Assignment 3: (2 marks)

Given below are some hints. For each O form, determine the E form (written directly as an English word, without an AI form).

-  is used after numbers in Science and can be used in place of  .
- A  is too much, but a  is nothing.
-  is a part of the name of a country, and  to you! This is the end.

– Mukund Choudhary

Problem 6: Altai-ed up

Here are some sentences in English and their corresponding Altaian[5] translations[6]:

мен кече ол атды сарайда тапганм.	I found that horse in the barn yesterday.
мен сенди айылда сакдым.	I waited for you at home.
бис ӧлӧнде ойнодыыс.	We played on the lawn.
ол кече бу самарады алган.	He/she received this letter yesterday.
олор кече кыр барган.	They went to a mountain yesterday.
бис кече ӱредичи школыпда сакганыс.	We waited for a teacher in school yesterday.
сен кече jӱсгенҥ.	You swam yesterday.
ӱредичи самара бичди.	The teacher wrote a letter.
зне кече бу бичикди кычырган.	The mother read this book yesterday.
уул кӧлди барды.	The boy went to the lake.
сен малды азырадын.	You fed the cattle.
бала самара турада тапды.	The child found a letter in the house.

[5]Altai is a Turkic language, spoken officially in the Altai Republic, Russia.

[6]а, о, у, ы are back/hard vowels and ӧ, ӱ, е, и are front/soft vowels. All other characters are consonants. Data has been simplified.

Assignment 1: (8 marks)

Translate the following English sentences to Altai:

(a) The mother found this book in the school.

(b) We played on the mountain yesterday.

(c) The boy fed a horse on the lawn.

(d) You wrote a letter yesterday.

Assignment 2: (8 marks)

Translate the following Altaian sentences to English:

(a) ол ол самарады кычырды.

(b) бала менди сакды.

(c) олор кече көлде jӱсген.

(d) мен кече школыпды барганм.

Assignment 3: (4 marks)

How are words or parts of words put together to form a valid Altaian sentence? What are their meanings and functions?

– Aparna Gupte

Problem No. 3 (20 Points):
Bariai

Consider the **unordered** squares of the numbers from 1 to 10 written out in the Bariai language[1]:

sangaul pange igegea lima ga pange
sangaul igegea lima ga ede
buno
sangalima sangaul ede igegea pange
pange
sangaul rua igegea lima
sangaul tol igegea lima ga ede
ede
lima ga pange
sangalima sangaul tol igegea ede

Assignment 1. Determine the values of all the squares given above.

Assignment 2. Write the following equality in figures:

sangaul pange igegea tol + sangaul igegea pange = sangalima igegea lima ga rua

Assignment 3. Spell out the following numerals in Bariai: 62; 88.

[1] The Bariai language belongs to the Oceanic branch of the Austronesian language family. It is spoken by more than 1300 people living in the western part of New Britain (Papua New Guinea).

- D. Gerasimov

Problem 5: Bahinemo

Here are the first ten multiples of a number less than 10 written in Bahinemo[4] in ascending order:

(a) həl baɸú həl ɸənbí husí

(b) həl baɸú həl baɸú wá ɸənbí gədà

(c) ímà tʰú hwəsə́ dəbátʰà

(d) ímà tʰú hwəsə́ yemú həl baɸú həl ɸənbí huməlí

(e) ímà tʰú hwəsə́ yemú həl baɸú həl baɸú wá baɸú

(f) ímà tʰú hwəsə́ husí husí

(g) ímà tʰú hwəsə́ husí həl baɸú həl ɸənbí gədà

(h) ímà tʰú hwəsə́ husí həl baɸú həl baɸú wá baɸú wá ɸənbí dəbátʰà

(i) ímà tʰú hwəsə́ huməlí huməlí

(j) ímà tʰú hwəsə́ huməlí həl baɸú həl baɸú

Bahinemo speakers often use a shortened form for some numbers. This, however, can lead to some ambiguity: a Bahinemo numeral in its shortened form can be interpreted as two different numbers.

Assignment 1: (4 marks)

Exactly one of the numbers given above is in its shortened form. All the other numbers are in the long form. In this case, the shortened numeral has only one interpretation, so there is no ambiguity. Which number is this, and what is its Hindu-Arabic equivalent? What is its Bahinemo long form? How is the shorthand form of a numeral derived from the long form?

Assignment 2: (8 marks)

Give all possible translations of the following Bahinemo numerals to Hindu-Arabic numbers. Some of these numerals may be in an ambiguous shortened form.

(a) ímà tʰú hwəsə́

(b) ímà tʰú hwəsə́ həl baɸú həl ɸənbí huməlí

(c) həl baɸú həl ɸənbí gədà

(d) ímà tʰú hwəsə́ həl baɸú həl baɸú həl baɸú həl baɸú wá ɸənbí husí

(e) ímà tʰú hwəsə́ həl baɸú

[4]Bahinemo is a Sepik language spoken in East Sepik Province, Papua-New Guinea. It is spoken in 4 villages, including in Gahom village of Tunap-Hunstein Rural LLG in East Sepik Province. Traditionally, Bahinemo speaking people rarely used large numbers in their daily life. Numbers above one hundred could in theory be correctly spoken, but the resulting phrase is so long and convoluted that hearers in the 1970s puzzled over them.

Assignment 3: (8 marks)

Translate the following Hindu-Arabic numerals to Bahinemo. When possible, give both the shorthand and the long form of the numbers in Bahinemo.

(a) 17

(b) 29

(c) 81

(d) 180

(e) 23

– Aparna Gupte